ARTIST TALK
WIND FARM, A COMPOSITION FOR LAPTOP ORCHESTRA

John Gibson
Indiana University Jacobs School of Music
Department of Composition

ABSTRACT

The author discusses *Wind Farm*, his recent composition for a laptop orchestra of at least seven players. The piece features two sound-creation interfaces, implemented in MaxMSP, and conductor software that determines some aspects of the sound made by individual players. The conductor and players communicate over a wireless network. The gestural control interface is simple, requiring only the built-in trackpad and computer keyboard. A Jitter patch provides video accompaniment.

1. INTRODUCTION

Wind Farm, a composition for laptop orchestra, was written at the request of the Electric Monster Ensemble at Montana State University, under the direction of Hsiao-Lan Wang. The piece touches metaphorically on the promise and problems of wind energy. Wind turbines are an important source of renewable energy, but the turbines kill birds, disrupt habitats, confuse airplane and weather radar, and spoil natural views. The members of the ensemble are drawn together in a common musical effort that represents the hope that at least some of these problems will be solved.

The *Wind Farm* software comprises three performance interfaces, implemented in MaxMSP: a filtered noise instrument (*blow*), suggestive of the wind; a resonant click instrument (*spin*), reminiscent of the sound of pinwheels or spinning wheels; and a conductor patch, which plays a drone during the run-up to the high point of the piece, but otherwise sends messages to the other players over an ad-hoc wireless network.

The music is a structured improvisation. The conductor navigates a set of textural cues and pitch collections to guide the players, who focus on gestures and timbral shaping.

2. IMPLEMENTATION

2.1. Low-key Conducting

Many laptop orchestras, such as the pioneering Princeton Laptop Orchestra (PLOrk), often employ a human conductor to lead the ensemble [1]. Although these conductors have developed new ways of communicating with players (for example, via hand signals), they retain a strong sense of hierarchical organization, so familiar from traditional Western orchestral practice, that might not be desirable for this new kind of music-making.

The conductor of *Wind Farm* maintains a much lower profile, both literally and figuratively. The *conductor* patch establishes a TCP/IP network connection with each of the players in the ensemble. Then she walks through a cue list. A *cue* here serves two functions: it displays a message for the player that briefly describes the texture and density to achieve, and it triggers instantaneous and gradual parameter changes. The conductor also moves through a set of pitch collections—one for each group of players, *blow* and *spin*. The pitches are distributed to the players in a way that scales transparently to the number of available musicians. The timing of cue and pitch changes is completely under the control of the person conducting, and so her job is to listen carefully to the players' improvised gestures and determine the best times to request changes. All of this guidance happens in a visually inconspicuous manner.

2.2. Spinning

The two sound-producing patches, *spin* and *blow*, let the players control their sound within the pitch and parameter constraints provided by the conductor. In both cases, the musician uses the built-in trackpad and keyboard of the laptop to shape the sound.

The *spin* patch implements *inertial scrolling*, familiar from the iPhone and similar devices, to allow the musician to generate a decelerating series of clicks. While your finger is on the trackpad, you can create individual clicks and short, fast streams of clicks. If your finger leaves the trackpad with enough momentum, the clicks continue a while before slowing to a stop. The inertial scrolling is implemented by feeding cursor position deltas from the Max *hi* (human interface device) object to a Javascript port of the FlickDynamics Objective C code, graciously offered under the BSD license by Dave Peck [2]. (Unfortunately, this seems to work only on Mac OS X; the built-in trackpads on Windows laptops this author has tried are not visible to the *hi* object.)

The clicks so generated run through several processing modules before leaving the patch: a delay-based resonator, a resonant low-pass filter, and a recirculating spectral delay. The beginning and ending of the piece present a monochromatic sound world, with only a weak sense of pitch, in contrast to the more colourful, strongly pitched sense of the rest of the music. The pitches are imposed by the resonator, while low-pass filter cutoff frequency sweeps, controlled by a simple computer keyboard scheme, impart dramatic timbral

contrasts between the sounds of the individual *spin* players. The spectral delay relies on the author's *jg.spectdelay~* external object, with source code available under the General Public License (GPL) [3].

2.3. Blowing

The *blow* patch lets the musician create a variety of filtered noise and other sounds by dragging a finger across the built-in trackpad, but without the inertial scrolling implemented in the *spin* patch. Although the trackpad is only a two-dimensional device (assuming no multi-touch capabilities), the *blow* patch divides an on-screen target rectangle into several regions that enable multi-dimensional control (see Figure 1). For example, dragging to the left of centre produces filtered noise with a relatively narrow bandwidth, while dragging to the right triggers a sustained, clear pitch, generated by the author's port of the BlowBottle physical modelling instrument from the Synthesis Toolkit (STK), by Perry Cook and Gary Scavone [4]. Near the climactic part of the piece, a sparse granular synthesis texture joins these other sounds, controlled by passing your finger through other regions of the target rectangle.

The player also operates a simple tremolo processor using the computer keyboard, while she controls overall volume with the Y axis of the trackpad.

3. PERFORMANCE CONSIDERATIONS

3.1. Spatialization

The *blow* and *spin* patches generate monophonic audio output, intended for reproduction by hemispherical loudspeakers. This type of loudspeaker is well suited to laptop ensemble performance, because it disperses sound in a much wider pattern than does a conventional loudspeaker, leading to a more natural, less focused sound, and making it easier for ensemble members to hear each other without using separate monitor speakers [5]. Placing the hemisphere as close as possible to the performer makes it more clear to the audience who is producing what sound, a consideration especially important for a medium in which there can be no fixed expectation of the kind of sound produced by a given musician. Placing players and speakers around the stage area with a reasonable amount of separation yields a vibrant, open sound with a great deal of spatial interest.

One drawback of the typical hemispherical loudspeaker is its lack of bass response. The *blow* and *spin* patches were designed to produce sound well above 100 Hz. Although the conductor plays only a single drone sound during one part of the piece, she has the distinction of generating both full-range audio, including a very deep bass that the other musicians cannot produce, and stereo sound. For this reason, the conductor requires a full-range stereo sound system, preferably equipped with a subwoofer.

3.2. What to Watch

For anyone witnessing a laptop performance, it can be hard to dismiss the nagging thought that the performers are merely checking their email. They sit motionless and stare at screens, oddly detached from the lively sounds they produce. In *Wind Farm*, a *spin* performer can alleviate this feeling somewhat by exaggerating the physical production of his inertial scrolling trackpad gestures, by moving his arm more than is really necessary. If he sits with the laptop tilted so that the audience can see the trackpad, it makes it easier for an audience member to understand what is happening. But there is probably no effective way for the *blow* player to communicate his actions to the audience, so small and precise are the finger movements required.

Partly for this reason, *Wind Farm* includes a Jitter patch that allows for live processing and compositing of wind turbine video loops during the performance.

3.3. Listening

The musical structure of *Wind Farm* encourages the players to listen to each other. In the quieter moments of the piece, when one person spins, the others should leave enough room for the gesture to speak. When a *blow* player sweeps the filter in a particular way, the others can complement this move. Because a player is not in control of her pitches, at any time she might receive notification that a new pitch has arrived from the conductor. Then she should complete her current gesture before starting another with the new pitch, all the while listening for the right moment to make the change.

It is satisfying that basic musical instincts and values can find expression in a performance context as technologically mediated as the laptop orchestra.

3.4. Performance Materials

Patches and performance information for *Wind Farm* are available at http://john-gibson.com/pieces/windfarm.htm.

4. REFERENCES

[1] http://plork.cs.princeton.edu (accessed 6 May, 2010).

[2] http://gist.github.com/100855 (accessed 6 May, 2010).

[3] Gibson, J. "Spectral Delay as a Compositional Resource," in *eContact!* vol. 11.4, *Toronto Electroacoustic Symposium 2009*, Canadian Electroacoustic Community, October 2009.

[4] http://ccrma.stanford.edu/software/stk (accessed 6 May, 2010)

[5] Trueman, D., Bahn, C., Cook, P. "Alternative Voices for Electronic Sound", in *Proceedings of the International Computer Music Conference*, Berlin, Germany, 2000.

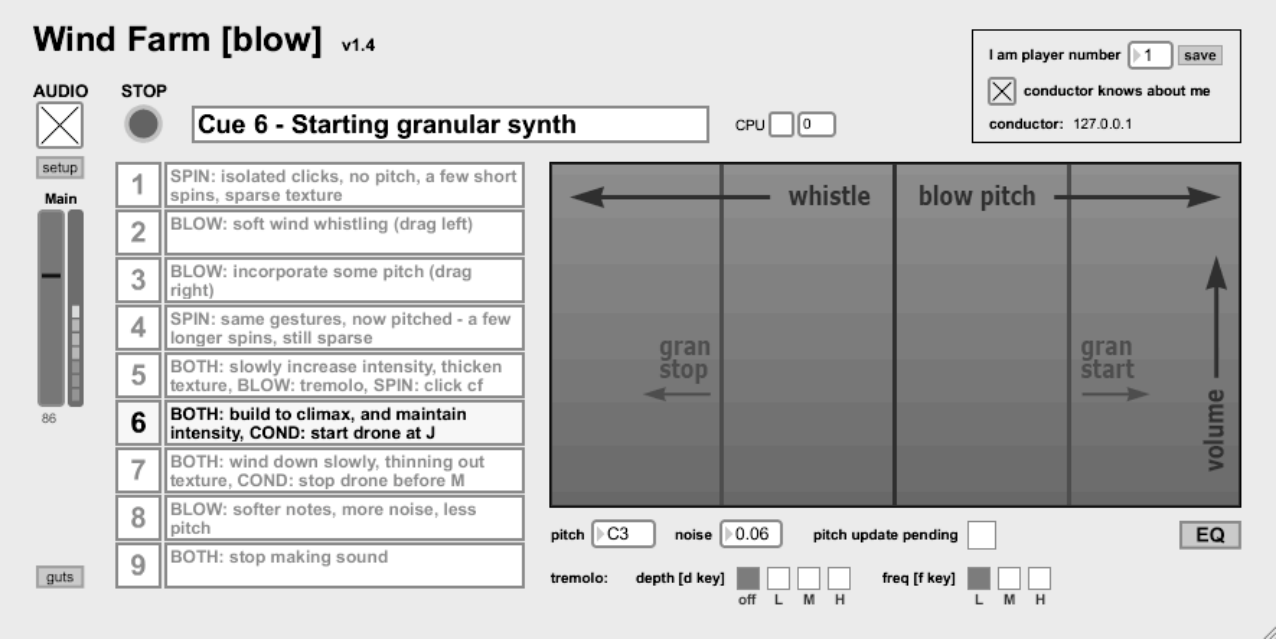


Figure 1. The *blow* patch from *Wind Farm*.